

TechNova Customer Support Bot

Project Evaluation and Performance Report

1. Project Overview

TechNova Customer Support Bot is an AI-powered customer support application for an electronics store. The system can answer questions about order status, returns, warranty, and shipping policies.

The application uses LangGraph to control the workflow, NVIDIA's hosted GPT-OSS-20B model for natural-language understanding, and custom tools for retrieving order and policy information.

2. Technology Stack

Component	Technology
Programming Language	Python
LLM	GPT-OSS-20B via NVIDIA
Workflow	LangGraph
LLM Framework	LangChain
Memory	LangGraph MemorySaver
Observability	LangSmith
Version Control	Git / GitHub

3. System Architecture

The application follows a tool-using agent workflow:

User Question → LangGraph Agent → GPT-OSS-20B → Tool Decision → Order/Policy Tool → Tool Result → LLM Response → User

The order-status tool retrieves information from the application's order data, while the policy tool retrieves returns, warranty, and shipping information.

4. Available Tools

Tool	Purpose
get_order_status	Retrieves order item, status, and ETA.
get_policy	Retrieves returns, warranty, and shipping policies.

5. Latency Measurement

The average measured latency across four test questions was **3.80 seconds**.

Question	Latency (seconds)
Where is my order TN1001?	2.94
What is the return policy?	5.8

What is the warranty policy?	2.63
What is the shipping policy?	3.82
Average	3.80

6. Token Usage

The average total token usage across the four test questions was **420.25 tokens per request**.

Question	Input	Output	Total
Where is my order TN1001?	373	48	421
What is the return policy?	366	61	427
What is the warranty policy?	368	40	408
What is the shipping policy?	372	53	425

7. Fault Testing

The chatbot successfully completed **4 of 4** fault-handling test cases (100%).

Test Case	Status
Unknown Order ID	PASS
Invalid Policy Topic	PASS
Empty Input	PASS
Unexpected Question	PASS

8. Functional Evaluation

The evaluation contained 5 predefined test cases. 5 tests passed, resulting in a measured accuracy of **100.00%** on this test set.

Test Case	Status
Order Status	PASS
Return Policy	PASS
Warranty Policy	PASS
Shipping Policy	PASS
Unknown Order	PASS

9. LangSmith Observability

LangSmith was configured to trace the chatbot workflow. The traces allow inspection of model calls, tool execution, runs, metadata, and overall execution flow. This provides useful visibility when debugging and evaluating the application.

10. Limitations

- The order and policy data are static demonstration data.
- Latency depends on external model/API response time.

- The functional evaluation uses a small predefined test set.
- The current chatbot can answer questions outside the intended customer-support domain.
- The measured evaluation accuracy should not be interpreted as general real-world accuracy.

11. Conclusion

The TechNova Customer Support Bot demonstrates a complete tool-using AI workflow using LangGraph, LangChain, and GPT-OSS-20B through NVIDIA. The project includes tool integration, conversation memory, observability, latency measurement, token tracking, fault testing, and functional evaluation.

The experiment results provide a reproducible baseline for future improvements such as stricter domain control, larger evaluation datasets, improved error handling, and performance optimization.